EuroMCM 2026 Problem α : Our Turn



A human skull versus a Neanderthal skull. Photo: Wikimedia Commons.

1 Track Notes

Problem α is the Continuous problem, though it also touches on the content of the Discrete (β), Sustainability (Ω) and Policy (θ) tracks. The track label describes the problem, not the methods a team is expected to use: solutions that draw on methods associated with any EuroMCM track are equally acceptable, and the choice of approach has no bearing on how a paper is judged. Creative solutions are encouraged.

2 Background

About 40,000 years ago the Neanderthals disappeared from Europe, within a few thousand years of the arrival of modern humans. When a newcomer that is better at capturing resources enters an occupied habitat, the species that dominated before can meet one of two fates. It may go extinct, or it may be pushed into a marginal niche and live on what the newcomer leaves.

Humans dominate through intelligence rather than strength, a position sometimes called the cognitive niche. Some researchers argue that AI systems able to improve their own design could one day enter that niche as newcomers. This process is called recursive self-improvement (RSI). Whether such systems will ever be built is unknown, and this problem does not ask you to judge that. If they are, they will still need electricity, hardware and land, and at first they will rely on human-run power plants and factories. Over a century, laws and markets may change beyond recognition, but at any moment only so much energy and land will be available. What matters is how much of them humans keep, however that is decided, not what happens to wages or jobs.

The Endowment for the Non-Obsolescence and Continuity of Humanity (ENOCH) has asked your team a conditional question. Suppose RSI systems exist. Treated as a problem in population dynamics, which fate awaits humans, and what does it depend on?

3 Requirements

Develop a model of a human population that shares the world with a population of RSI systems. You may treat AI capability as a single quantity, and you need no knowledge of machine learning.

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- Treat RSI systems as a new kind of organism: estimate what they consume to exist and expand (energy, hardware, land), how fast they grow, and what physically limits them.
- Let humans and RSI systems share land and energy. Test this shared part on Neanderthals and modern humans: with parameters taken from the energy and density figures in Section 4, does it give a coexistence time of the observed order of magnitude? Say which features of that case carry over to RSI. Then add what RSI systems first need from humans, such as power plants and chip factories, and how that need changes as they improve.
- Decide how alignment acts in your model and treat it as a setting you vary, not a quantity to predict. Express it on a scale from 0, where RSI systems ignore human needs, to 1, where they take nothing humans need.
- Define extinction, marginal survival and a time horizon in your own terms, estimate how likely each fate is, and find how much alignment humans need to avoid both.
- Compare at least two actions humans could take, such as capping computing power, a shutdown mechanism that can fail, or enhancing human abilities, and show how their timing changes the outcome.

No forecast here can be checked. Your work will be judged on how well each assumption is tied to a physical quantity or a documented case, and on your analysis of what the result depends on, not on the numbers themselves.

4 Data

Published figures on the Neanderthal replacement are ranges rather than time series: good for checking a model's order of magnitude, not for fitting curves. Some are disputed, and not all describe the same region.

- Coexistence of Neanderthals and modern humans in Europe: 2,600–5,400 years (Higham et al. 2014).
- Coexistence in France and northern Spain: 1,400–2,900 years (Djakovic et al. 2022).
- Population density of modern humans relative to Neanderthals in south-west France: about 10 to 1, by a disputed method (Mellars and French 2011).
- Daily energy use of an adult Neanderthal: 3,000–5,000 kcal for women and 4,000–7,000 kcal for men (Snodgrass and Leonard 2009).
- Extra daily energy a Neanderthal needs over a modern human in the same climate: 100–350 kcal (Froehle and Churchill 2009).

For scale, data centres used about 415 TWh of electricity in 2024, 1.5% of the world total. Their use has grown about 12% a year since 2017 and is expected to reach about 945 TWh by 2030 (IEA, Energy and AI, 2025). External data may be used if sources and assumptions are stated.

5 Share Your Insights

Write a one-page letter telling the ENOCH board what your model says, how much confidence it deserves, and which unknown quantity is most worth measuring. Name one present-day question your model could also help answer.

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6 Submission

Your PDF solution (≤25 pages) should include:

- One-page Summary Sheet
- Table of Contents
- Complete Solution
- One-page Letter
- References
- Report on Use of AI Tools (if applicable; excluded from the 25-page limit)

There is no mandatory minimum length. Teams may submit incomplete solutions. The use of AI tools is allowed but optional; compliance with EuroMCM AI usage policy is required.

7 Links

Speculations concerning the first ultraintelligent machine

Anthropic's Responsible Scaling Policy

A mathematical model for Neanderthal extinction

An ecocultural model predicts Neanderthal extinction through competition with modern humans

A parsimonious neutral model suggests Neanderthal replacement was determined by migration and random species drift

8 Glossary

Recursive self-improvement (RSI): A process in which an AI system improves its own design, and the improved system is then better at making further improvements.

Niche: The set of resources and conditions that a species needs in order to live and reproduce.

Competitive exclusion: The principle that two species limited by the same resource cannot coexist indefinitely, so that one displaces the other.

Carrying capacity: The largest population that the resources of an environment can sustain indefinitely.

Alignment: The degree to which the goals of an AI system take human needs into account.

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